Notes on Algebraic Geometry

Riccardo Lo Iacono

March 24, 2026

Document is WIP, typos could be found

Contents.

1 Introduction and Algebra reminders **1**
1.1 Algebra reminders 1
1.2 Noetherian rings 3

2 Affine algebraic sets **6**
2.1 Noetherian topological spaces . . . 9

- 1 - Introduction and Algebra reminders.

Among the many problems that algebraic geometry tries to solve, there is that of solving systems of homogeneous polynomials. That is, we'd like to use the tools provided by algebra and geometry to study the zeros of polynomials. Meaning we'd like to solve systems of the form

$$\begin{cases} f_1(x_1, x_2, \dots, x_m) = 0 \\ f_2(x_1, x_2, \dots, x_m) = 0 \\ \quad\vdots \qquad \vdots \\ f_n(x_1, x_2, \dots, x_m) = 0 \end{cases}$$

where all $f_i, i = 1, 2, \dots, n$ are either homogeneous polynomials over some projective space, or simply polynomials.

We now introduce one of the fundamental results on the topic, which will be proven later on.

Theorem 1.1 *Let $\mathbb{F}$ be a field algebrically closed. Then, a homogeneous system of n equations admits non-trivial solutions if and only if $n \leq m$, m being the number of variables in each polynomial.*

Another issue is that of describing the geometry of such solutions. In this context, the following classification is available.

- $\mathbb{F}$ is an algebrically closed field:
  - **dim 1:** the result $\mathcal{C} \subseteq \mathbb{A}^2$ is generalized.
  - **dim 2:** several results by *Castelnovo, Eriques, Severi, etc.*
  - **dim $\geq$ 3:** many open questions.
- $\mathbb{F}$ is not an algebrically closed field:
  - Specific restrictions. For instance, the arithmetic algebric geometry used to prove Fermat's theorem ($x^n + y^n = z^n$).

- 1.1 - Algebra reminders.

Throughout these notes we remind several concepts of both Algebra and Geometry. For the time being we focus on a brief Algebra reminder, in which we focus on some results on rings and ideals.

Definition (Rings) Let R be a set and

$$+ : R \times R \to R$$
$$* : R \times R \to R$$

be two operations. We say that R is a (commutative) ring if the following holds.

- For any $a, b, c \in R, a + (b + c) = (a + b) + c$.
- Exists an element $0 \in R$ such that $a + 0 = 0 + a = a$.
- For any $a \in R$, there is $-a \in R$ such that $a + (-a) = -a + a = 0$.
- For any $a, b \in R, a + b = b + a$.
- For any $a, b, c \in R, a * (b * c) = (a * b) * c$.
- For all $a, b \in R, a * b = b * a$.
- $*$ distributes over $+$.

Definition (Ideal) Let R be a ring. We say that a subset $I \subseteq R$ is and ideal of R if it holds that

- I is subgroup for $(R, +)$. That is, if
$$\forall x, y \in I, x + y \in I.$$
- $\forall a \in R, \forall x \in I, a * x \in I$.

Remark. From now on we write ab instead of $a * b$.

Definition. Let R be a ring and let $T \subseteq R$ be a subset. We define the set
$$(T) = \left\{ \sum_{i=0}^{n} a_i x_i \;\middle|\; a_i \in R, x_i \in T \right\}.$$

Additionally, if $I = (T)$ and $|T| < \infty$ we say that I is *finetely generated*.

Definition. Let R, Q be rings. A function $f : R \to Q$ is called (ring) homomorphism if

- $f(a + b) = f(a) + f(b), \forall a, b \in R$;
- $f(ab) = f(a)f(b), \forall a, b \in R$;
- $f(1_R) = 1_Q$.

Remark. Given $f : R \to Q$ a ring homomorphism, it can be proved that

$$\mathrm{Ker}(f) = \{a \in R \mid f(a) = 0\}$$

is an ideal of R. On the other hand, $\mathrm{Im}(f) = \{b \in Q \mid b = f(a), a \in R\}$ is a subring of Q.

Recall that, given R a ring a $I \subseteq R$ an ideal, we can consider the quotient ring

$$\frac{A}{I} = \{a + I \mid a \in R\}.$$

We usually write $\overline{a}$ instead of $a + I$.

Lastly, we present a classic result in ring theory,

Theorem (First isomorphism theorem) *Given a homomorphism $f : R \to Q$, there exists a unique homomorphism g such that $f = \pi \circ g$, where π is the canonical epimorphism. In terms of commutative diagrams, we have the following.*

$$\begin{tikzcd} R \arrow[r, "f"] \arrow[d, "\pi"] & f(R) \subseteq Q \\ \frac{R}{I} \arrow[ur, "g"] & \end{tikzcd}$$

Proposition. Given R a ring and $\mathfrak{p}$ an ideal, we say that $\mathfrak{p}$ is a prime ideal of R, if

$$ab \in \mathfrak{p} \implies a \in \mathfrak{p} \vee b \in \mathfrak{p}$$

or equivalently, if $a, b \notin \mathfrak{p} \implies ab \notin \mathfrak{p}$.

- 1.2 - Noetherian rings.

Definition. Let S be a set. We define a sequence of inclusion of the form

$$S_1 \subseteq S_2 \subseteq \cdots \subseteq S_k \subseteq \cdots$$

a (ascending) chain. Additionally, if there $\exists n \in \mathbb{N}$ such that $S_n = S_{n+1} = \cdots$, we say that the chain is stationary.

Definition (Noetherian ring) Let R be a ring. We say that R is Noetherian if and only if each ascending chain of ideals of R is stationary,

Proposition. Let R be a ring. Then, the following are equivalent.

1. R is Noetherian.
2. Each ideal I of R is finitely generated.

3. However took a family $\mathcal{F}$ of ideals of R; $\mathcal{F}$ admits maximum.

Proof

$2 \implies 1$**:** Let $I_1 \subseteq I_2 \subseteq \cdots \subseteq I_k \subseteq \cdots$ be a chain of ideals. Let

$$I = \bigcup_{j=1}^{\infty} I_j$$

be an ideal finitely generated of R. That is, $I = (x_1, x_2, \ldots, x_n)$.

By the way I is defined, this means that

$$\exists k_1, k_2, \ldots, k_n \text{ s.t. } x_i \in I_{k_i},$$

hence, if $m = \max\{k_i\}$ then $x_i \in I_m = I_{m+1} = \cdots$; that is, R is Noetherian.

$1 \implies 3$**:** Consider $\mathcal{F}$ be a non-empty family of ideals. Let $I_1 \in \mathcal{F}$. It follows that

- I_1 is maximal, from which the thesis.
- I_1 is not maximal. By hypothesis, there exists $I_2 \supseteq I_1$.

Iterating the procedure we either find a maximal ideal, from which the thesis; or we end up with a non-stationary chain, which is absurd.

$3 \implies 3$**:** Let $I \subseteq R$ be an ideal, $I \neq (0)$. Consider the following family of ideals

$$\mathcal{F} = \{J \subseteq R \mid J \text{ is finitely generated}, J \subseteq I\}.$$

Trivially, $\mathcal{F} \neq \varnothing$. Let I_{MAX} be the maximal element in $\mathcal{F}$. Then $I_{MAX} = (x_1, \ldots, x_m)$ with $x_i \in R$. Assume, by the sake of contradiction that $I_{MAX} \neq I$, this implies that $\exists y \in I$ such that $y \notin I_{MAX}$. But then we should have $(x_1, \ldots, x_m, y) \in \mathcal{F}$, meaning I_{MAX} isn't the maximal element, which is absurd. ■

Proposition. Let $f : R \to Q$ be a ring homomorphism. Then, if f is onto and R is a Noetherian ring, Q is a Noetherian ring.

Proof Let $J \subseteq Q$ be an ideal. By the properties of homomorphism

$$I = f^{-1}(J) = \{a \in R \mid f(a) \in J\}$$

is an ideal of R.

By hypothesis we have that R is Noetherian, then we can assume that $I = (a_1, a_2, \ldots, a_m)$. We prove that, $J = (b_1, b_2, \ldots, b_m)$, where $b_i = f(a_i)$.

Consider any $y \in J \implies \exists x \in I$ s.t. $f(x) = y$. But

$$x = \sum_{i=0}^{n} a_i x_i, x_i \in R.$$

Hence, we have that

$$\begin{aligned} y &= f\left(\sum_{i=1}^{n} a_i x_i\right) \\ &= \sum_{i=1}^{n} f(a_i x_i) \\ &= \sum_{i=1}^{n} \underbrace{f(a_i)}_{=b_i} \underbrace{f(x_i)}_{\in J} \end{aligned}$$

therefore $y \in (b_1, b_2, \ldots, b_n)$ and $J \subseteq (b_1, b_2, \ldots, b_n)$. Proving that $(b_1, b_2, \ldots, b_n) \subseteq J$ is trivial.

Theorem (Hilbert's base) *Let R be a ring. If R is Noetherian, then $R[x]$ is Noetherian.*

Corollary. Let $\mathbb{F}$ be a field. Then, $\mathbb{F}[x_1, x_2, \ldots, x_n]$ is Noetherian.

Proof Let us proceed by induction on the number of variables. By the previous theorem we know that for $n = 1, \mathbb{F}[x_1]$ is Noetherian. As inductive step, assume the statement to be true for $n-1$; that is, $\mathbb{F}[x_1, x_2, \ldots, x_{n-1}]$ is Noetherian. Then, we have that $\mathbb{F}[x_1, x_2, \ldots, x_{n-1}, x_n] = \mathbb{F}[x_1, x_2, \ldots, x_{n-1}][x_n]$, which is true by hypothesis. ■

Corollary. Given $I \subseteq \mathbb{F}[x_1, x_2, \ldots, x_n]$, the ring

$$B = \frac{\mathbb{F}[x_1, x_2, \ldots, x_n]}{I}$$

is Noetherian.

Proof Trivial. By assuming the epimorphism

$$\mathbb{F}[x_1, x_2, \ldots, x_n] \to \frac{\mathbb{F}[x_1, x_2, \ldots, x_n]}{I},$$

the statement is true by the previously discussed proposition. ■

Proposition. Let R be a Noetherian ring, and let $J \subseteq R$ be an ideal. Then, $\exists M \subseteq R$ maximal such that $J \subseteq M$.

Proof Simply consider the family

$$\mathcal{F} = \{I \subseteq R \mid J \subseteq I\}.$$

Since R is Noetherian, then $\mathcal{F}$ must admit a maximal element. ■

- 2 - Affine algebraic sets.

We begin by recalling the concepts of radical ideal. Let us also note that, in the remaining part of these notes, we will be considering Noetherian rings unless otherwise specified.

Definition. Let R be a ring and let $I \subseteq R$ be an ideal. The set

$$\sqrt{I} = \left\{ x \in R \,\middle|\, x^k \in I, k \in \mathbb{N} \right\}$$

Proposition. Given R a ring and $I \subseteq R$ an ideal, the set $\sqrt{I}$ is an ideal and $\sqrt{I} \supseteq I$.

Proof We have to prove that $\sqrt{I}$ is closed under addition and multiplication.

Addition: let $x, y \in \sqrt{I}$. Then, $\exists n, m \in \mathbb{N}$ such that $x^n, y^m \in I$. Hence, we have that

$$(x+y) \in \sqrt{I} \iff (x+y)^{m+n} \in I.$$

Observe that

$$(x+y)^{m+n} = \sum_{k=1}^{m+n} \binom{m+n}{k} x^{m+n-k} y^k.$$

from which

1. if $k \leq m$, then $x^{m+n-k} y^k = \underbrace{x^n}_{\in I} \underbrace{x^{m-k} y^k}_{\in R}$.
2. if $m+1 \leq k \leq m+n$, then $x^{m+n-k} y^k = \underbrace{y^m}_{\in I} \underbrace{x^{m+n-k} y^{k-m}}_{\in R}$.

Therefore, any term of the summation is in I. Thus, $(x+y)^{m+n} \in I$.

Multiplication: let $x \in \sqrt{I}, a \in R$. We have that

$$ax \in \sqrt{I} \iff (ax)^n \in I.$$

But

$$(ax)^n = a^n x^n \in I,$$

therefore $ax \in \sqrt{I}$. ■

Proposition. Given R a ring, and $I \subseteq R$ an ideal, it holds that

$$I \text{ is radical} \iff \frac{R}{I} \text{ is reduced.}$$

Proof Trivial. Simply consider the epimorphism $\pi : I \to \frac{R}{I}$. Note that $\mathrm{Ker}(\pi) = I$, from which the thesis follows easely. ■

Definition (Affine algebraic set)

1. Let $\mathbb{F}$ be a field and let $\mathcal{K} = \mathbb{F}[x_1, x_2, \ldots, x_n]$. Let $F(x_1, \ldots, x_n) \in \mathcal{K}$. If $(a_1, \ldots, a_n) \in \mathbb{F}^n$ is such that $F(a_1, \ldots, a_n) = 0$, we say that the point $P = (a_1, \ldots, a_n) \in \mathbb{A}^n$ is a zero for F.

2. Let $S \subseteq \mathcal{K}$ be a set of polynomials. The set

$$V(S) = \{(a_1, \ldots, a_n) \in \mathbb{F}^n \mid \forall F \in S, F(a_1, \ldots, a_n) = 0\}$$

is called affine algebraic set.

Proposition. Given $S \subseteq \mathcal{K} = \mathbb{F}[x_1, x_2, \ldots, x_n]$ a set of polynomials, the sets $V(S), V(I(S)), V(\{G_1, \ldots, G_n\})$ coincide.

Proof We prove that, if $I(S)$ is generated by S, then $V(S) = V(I(S))$. Consider $F \in I(S)$, we can write it as $F = \sum_{i=1}^{n} F_i a_i, F_i \in S$.

($\subseteq$) If $P \in V(S)$, then P is a zero for any polynomial in S; that is, $F(P) = 0$. Hence, $P \in V(I(S))$.

($\subseteq$) If $P \in V(I(S))$, then $F(P) = 0$ for any $F \in I$. By hypothesis I is generated by S, meaning $S \subseteq I$. Hence, $G(P) = 0, \forall G \in S$. ■

Proposition. Affine algebraic sets satisfy the properties of closed sets in a topology.

Proof What we have to prove is that, the union of finitely many affine sets, is an affine set; and likewise, that the intersection of finitely many affine sets is a affine set.

Let us consider the intersection first. Let $V(S_1), \ldots, V(S_k)$ be a finite sequence of affine sets. Let

$$V(S) = \bigcap_{i=1}^{k} V(S_i)$$

be the intersection of all $V(S_i)$. We prove that if $P \in V(S)$, then $P \in V(S_i), \forall i$.

By the definition of affine set

$$V(S) = \{P \in \mathbb{A}^n \mid f(P) = 0, \forall f \in S\},$$

meaning that

$$V(S) = \bigcap_{i=1}^{k} V(S_i) = V\left(\bigcup_{i=1}^{k} S_i\right).$$

For what regards the union, we can focus on the simple case of the union of two affine sets. Let $V_1 = V(F_1, \dots, F_k), V_2 = V(G_1, \dots, G_l)$ be affine sets. We prove that $V_1 \cup V_2 = V = V(F_1, \dots, F_k, G_1, \dots, G_l)$.

$\subseteq$: Let $P \in V_1 \cup V_2$. Then, $P \in V_1 \vee P \in V_2$. In any case $F_i(P)G_j(P) = 0$ for any i, j. Meaning $P \in V$.

$\supseteq$: Let $P \in V$. Then $F_i(P)G_j(P) = 0$ for any i, j. Meaning that

- If $F_i(P) = 0, P \in V_1$, or
- If $F_i(0) \neq 0 \implies G_j(P) = 0, P \in V_2$.

In any case $P \in V_1 \cup V_2$. ■

Definition (Zarinski topology) A topology of $\mathbb{A}^n$ is called a Zarinski topology if its closed sets are affine algebraic sets.

If $X \subseteq \mathbb{A}^n$ is a subset, its Zarinski topology is the one induced by the one of $\mathbb{A}^n$. Its closed sets have the shape

$$X \cap S(\mathcal{K}), \mathcal{K} = \mathbb{F}[x_1, \dots, x_n].$$

Definition. Let $V \subseteq \mathbb{A}^n$. The set

$$I(V) = \{f \in \mathcal{K} \mid f(P) = 0, \forall P \in V\}$$

is the set of those polynomials that go to zero in all points of V.

Remark. It can be proved that $I(V)$ is an ideal. Also, two maps can be defined.

$$\begin{aligned} \{J \mid J \text{ ideal of } \mathcal{K}\} &\xrightarrow{V} \{X \subseteq \mathbb{A}^n\} \\ J &\longmapsto V(J) \end{aligned}$$

$$\begin{aligned} \{X \subseteq \mathbb{A}^n\} &\xrightarrow{I} \{J \mid J \text{ ideal of } \mathcal{K}\} \\ X &\longmapsto I(X) \end{aligned}$$

Proposition. Let $X \subseteq \mathbb{A}^n$, then

1. $X \subseteq V(I(X))$;
2. $X = V(I(X)) \iff X$ closed set;
3. $I(X)$ radical ideal;
4. There exist ϕ one-to-one:

$$\{X \in \mathbb{A}^n \mid X \text{ closed set}\} \xrightarrow{\phi} \{I \subseteq \mathcal{K} \mid I \text{ radical ideal}\}$$

Proof

1. Trivially true. In fact, if $F(x_1, \dots, x_n) \in I(X)$ this means that $F(P) = 0, \forall P \in X$. Hence, a point $P \in X$ is such that $G(P) = 0, \forall G \in I(X)$. Thus, $\forall P \in X, P \in V(I(X))$.
2. Let us note that the left implication is true by definition. The other implication follows. Let $X = V(S)$ closed affine set. By definition, $S \subseteq I(X)$, which means $X \subseteq V(I(X)) \subseteq V(S)$. But we've said that $X = V(S)$; therefore, $V(I(X)) \subseteq X \implies X = V(I(X))$.
3. Let $F^n(x_1, \dots, x_n) \in I(X)$. Hence $F^n(P) = 0, \forall P \in X$. But note that $F^n(P) = (F(P))^n \implies (F(P))^n = 0 \implies F \in I(X)$.
4. We can simply apply the criterion for a function to be one-to-one. In fact, if X_1, X_2 are closed affine sets such that $I(X_1) = I(X_2)$, then trivially $V(I(X_1)) = V(I(X_2)) \implies X_1 = X_2$. ■

- 2.1 - Noetherian topological spaces.

Definition. Let X be a topological space. We say that X is Noetherian if and only if, any descending chain of closed sets

$$X = X_1 \supseteq X_2 \cdots \supseteq X_k \supseteq \cdots$$

is stationary.

Note that if $X \subseteq \mathbb{A}^n$, then X is a Noetherian topological space. We now recall what a reducible/inreducible space is.

Definition. Let X be a topological space. Then, X is reducible if $X = X_1 \cup X_2$ with X_1, X_2 proper closed subsets. Is inreducible viceversa.

Lemma. *Let $X = X_1 \cup \cdots \cup X_k$ be a topological space, where X_i is a proper closed set. Then, X is reducible.*

Proof Let us prove the result by induction on the number of closed sets.

$k = 2$**:** follows by the definition.

$k \geq 3$**:** Assume that for $k-1$ it holds that X is reducible. This means that $X = X_1 \cup \cdots \cup X_{k-1}$. But each X_i is a closed set; additionally, a finite union of closed sets is still a closed set. Hence, we have that

$$X = \underbrace{Z}_{X_1 \cup \cdots \cup X_{k-1}} \cup X_k.$$

Therefore X is reducible. ■

Lemma. *Let X be a topological space. Then, the following are equivalent.*

1. *X is inreducible.*
2. *For any $U_1, U_2 \subseteq X, U_i \neq \varnothing$, then $U_1 \cap U_2 \neq \varnothing$.*
3. *Each non-empty open set $U \subseteq X$ is dense in X.*

Proof

$1 \implies 2$**:** Let U_1, U_2 be non-empty open sets of X. We can write

$$U_1 = X \setminus X_1 \text{ and } \quad U_2 = X \setminus X_2,$$

with X_1, X_2 proper closed sets of X. By the sake of contradiction, let $U_1 \cap U_2 = \varnothing$. Then we have that

$$\begin{aligned} U_1 \cap U_2 &= X \setminus X_1 \cap X \setminus X_2 \\ &= X \setminus (X_1 \cup X_2) = \varnothing. \end{aligned}$$

Thus we have $X = X_1 \cup X_2$, which is a contradiction.

$2 \implies 3$**:** Let $U \subseteq X$ be an non-empty open set. Then $\forall W \subseteq X$, such that W is open and non-empty, we have that $U \cap W \neq \varnothing$. By the criterion for density, it follows that U is dense.

$3 \implies 1$**:** Let check this one. ■

Theorem. *Let $V \subseteq \mathbb{A}^n$ be an affine closed set. Then, V is inreducible if and only if $I(V)$ is a prime ideal.*

Proof

$\Longleftarrow$**:** By the sake of contradiction let V be reducible. Then, $V = V_1 \cup V_2$, with V_i be proper closed sets. We have that

$$V_1 \subset V \text{ and } V_2 \subset X.$$

Considering the ideals, we have

$$I(V_1) \supset I(V) \text{ and } I(V_2) \supset I(V).$$

Hence, there exist $F_1, F_2 \in \mathcal{K}$ such that $F_1 \in I(V_1), F_2 \in I(V_2), F_1, F_2 \neq I(V)$. Thus, $\forall P \in V, F_1F_2(P) = F_1(P)F_2(P) = 0$. Meaning $P \in V_1 \vee P \in V_2 \implies F_1F_2 \in I(V)$ which is a contradiction.

$\Longrightarrow$**:** Assume that $I(V)$ is not prime. Therefore, $\exists F_1, F_2 \in \mathcal{K}, F_1, F_2 \notin I(V) \wedge F_1F_2 \in I(V)$.

Let $S_i = I(V) \cup \{F_i\}$. We have

$$\begin{aligned} V(S_i) &= V(\{F_i\} \cup I(V)) \\ &= V(\{F_i\}) \cap V, i = 1, 2 \\ &= \{P \in V \mid F_i(P) = 0\}. \end{aligned}$$

Thus, each V_i is a proper closed subset of V.

We prove that $V_1 \cup V_2 = V$. Let $P \in V$. Then, $F_1F_2(P) = 0 \implies P \in V_1 \vee P \in V_2$, hence $P \in V_1 \cup V_2$. ■

Remark. Note that if $\mathbb{F}$ is an infinite field, then $\mathbb{A}^n$ is inreducible.
In fact, if $V = \mathbb{A}^n$, Considering the ideals this generate we have

$$I(V) = I(\mathbb{A}^n) = \{F \in \mathcal{K}[x_1, \ldots, x_n] \mid F(P) = 0, \forall P \in \mathbb{A}^n\}.$$

By the identity principle , we have that the only polynomial in $I(V)$ is the null one; hence $I(V)$ is an integral domain, thus is prime and therefore V is inreducible.

ToDo. Search what this "indentity principle" is.

Proposition. Noetherian topological spaces satisfy the following.

1. If X is a Noetherian topological space and $Y \subseteq X$ is a topological subspace, then Y is Noetherian.
2. The union of finitely many Noetherian topological spaces is Noetherian.
3. If X is a Noetherian topological space, then X is semi-compact.

Proof

1. Note that we can write Y as a chain of closed sets; that is, $Y = Y_1 \supseteq Y_2 \cdots \supseteq Y_k \supseteq \cdots$, where each $Y_i = Y \cap X_i$. What we are left with is proving that the X_is form a chain in X. Let us rewrite the X_i.

   Let $X_1' = X_1$. For any $i \geq 2$ we have $Y_i = Y \cap X_i$ and $Y_i \subseteq Y_{i-1}$, meaning that $Y_i \subseteq X_{i-1}$; that is, $Y_i = Y \cap (X_i \cap \cdots \cap X_1) = Y \cap X_i'$.

   From the above $X_1' \subseteq X_2' \subseteq \cdots \subseteq X_k' \subseteq \cdots$. Since X is Noetherian, the chain of X_i' must be stationary, hence the chain of Y_i is also stationary.

2. Since we are considering a finite union, we can simply prove the proposition for $n = 2$, as the proof for $n \geq 3$ follows by induction.

   Let Y, Z be Noetherian spaces. We prove that $Y \cup Z = X$ is a Noetherian spaces.

   Since Y, Z are Noetherian we know that

$$\begin{aligned} Y &= Y_1 \supseteq Y_2 \supseteq \cdots \supseteq Y_k \supseteq \cdots \\ Z &= Z_1 \supseteq Z_2 \supseteq \cdots \supseteq Z_l \supseteq \cdots . \end{aligned}$$

   Additionally, each Y_i and Z_i can be seen as the intersection of some closed set of X with Y and Z respectively. That is, $Y_i = Y \cap X_i$ and $Z_i = Z \cap X_i$.

   Now we know that $\exists s \in \mathbb{N}$ such that $Y_s = Y_{s+1} = \cdots$ and $\exists t \in \mathbb{N}$ such that $Z_t = Z_{t+1} = \cdots$. Hence, if $m = \max\{s, t\}$ we have that $Y_m = Y_{Y+1} = \cdots$ and $Z_m = Z_{m+1} = \cdots$. But for all i, it holds

$$X_i = (Y \cap X_i) \cup (Z \cap X_i) = (Y \cup Z) \cap X_i$$

   by which $X_m = X_{m+1} = \cdots$, meaning that X is Noetherian.

3. Let $X = \bigcup_{i \in I} U_i$, with I set of indexes.

   Let $i_1 \in I$. If $X = U_{i_1}$ we are done. Otherwise, there exists some i_2 such that $U_{i_2} \subset U_{i_2}$. If $U_{i_1} \cup U_{i_2} = X$ we are done, otherwise we repeat the process.

   Two scenarios follow:

   (a) $\exists i_n \in I$ such that $U_{i_1} \cup U_{i_2} \cup \cdots \cup U_{i_n} = X$. In which case we get the thesis; or

   (b) We get a chain of open sets that is not stationary. Meaning that, once we consider the complements, we have a chain of closed sets that is not stationary, which contradicts the hypotesis of X being Noetherian. ■

Lemma. *Let X be a Noetherian space. If $\mathcal{F} \neq \varnothing$ is a family of closed sets, then exists $m \in \mathcal{F}$ minimum.*

Proof Trivial. Analogous to the proof of the existence of a maximal element in a Noetherian ring. ■

- 2.1.1 - Unique decomposition.

Definition. Let X be a topological space. We say that X admits a decomposition in inreducible closed sets, if there exists $X_1, \dots, X_n$ inreducible closed sets, such that $X = X_1 \cup \dots \cup X_n$.

Lemma. *Let X be a topological space. We say that a decomposition $X = X_1 \cup \dots \cup X_n$ is irredundant if any of the following holds.*

1. *For all i, it holds $X_i \not\subset \bigcup_{j \neq i} X_j$.*
2. *$\forall i \neq j, X_i \not\subset X_j$.*

Proof We prove that the two statements are equivalent.

$1 \implies 2$**:** By the sake of contradiction, let 2 be false. Then, there exists i such that $X_i \subseteq X_j$, meaning

$$X_i \subseteq \bigcup_{j \neq i} X_j.$$

Which contradicts the hypotesis.

$2 \implies 1$**:** By a similar reasoning as above, if 1 does not hold, then there exists i such that $X_i \subseteq \bigcup_{j \neq i} X_j$.

Hence, $X_i = \bigcup_{j \neq i} (X_j \cap X_i) = \bigcup_{j \neq i} Y_j$. Each Y_i is closed in X_i but not proper since the X_i are inreducible. Therefore, there exists $j \neq i$ such that $Y_j = X_i$; that is, $X_i \cap X_j = X_i \iff X_i \subseteq X_j$. ■

Theorem. *Let X be a Noetherian space. Then, X admits an irredundant decomposition in inreducible closed sets. Additionally, such decomposition is unique apart from the ordering.*

Proof We divide the proof in three parts: we first show that a decomposition exists, then we prove that it's irredundant and lastly we show its uniqueness.

1. Let $\mathcal{F}$ be a family of closed sets in X that admit no decomposition. We show that $\mathcal{F} = \varnothing$.

   By the sake of contradiction assume $\mathbb{F} \neq \varnothing$. Then there exists $Z \in \mathcal{F}$, such that Z is minimal. We have that

(a) If Z is inreducible, then $Z = Z$ is a decomposition in inreducible closed sets. Which goes against the hypotesis.

(b) If Z is reducible, then we can write $Z = Z_1 \cup \cdots Z_k$, with each Z_i a proper closed set. Hence, $Z_i \neq \mathcal{F}$; that is, Z_i admits a decomposition, which is again a contradiction.

2. Trivial. Let $X = X_1 \cup \cdots \cup X_k$ be a decomposition with a minimal element. By the above lemma, $\forall i, X_i \not\subset \bigcup_{j \neq i} X_j$.

3. Let $X = X_1 \cup X_2 \cup \cdots \cup X_k$ and $X = Y_1 \cup Y_2 \cup \cdots \cup Y_l$ be two decomposition for X, with X_i, Y_j inreducible closed sets. We have

$$\begin{aligned} X_i &= X_i \cap X \\ &= X_i \cap (Y_1 \cup Y_2 \cap \cdots \cap Y_l) \\ &= \bigcup_{j \neq i} (Y_j \cap X_i). \end{aligned}$$

But we said that each X_i is inreducible, hence $\exists j = j(i)$, where by $j(i)$ that j depends from i, such that $X_i \cap Y_{j(i)} = X_i$; that is, such that $X_i \subseteq Y_{j(i)}$.

Applying the same reasoning for the Y_js, we get

$$X_i \subseteq Y_{j(i)} \subseteq X_{i(j(i))}.$$

Hence, $X_i = Y_{j(i)}$. ■